Battery Parameters

To be measured:

- Voltage
- Current
- Temperature

To be calculated:

- Energy
- Power
- Open Circuit Voltage (OCV)
- SoC
- SoH
- DoD (Depth of Discharge)
- C rating
- Cycle counting
- Internal resistance

To be graphed:

- SoC vs time
- SoH vs time or cycle
- Temperature vs time
- Internal resistance vs time or cycle
- Charge-Discharge efficiency curve
- Voltage vs SoC Curve

To be predicted:

- Remaining Useful Time (RUL)
- Capacity Fade Rate
- Estimate battery lifetime
- Aging
- Self-discharge rate
- Fault prediction using voltage sag
- Battery Health Index to predict next maintenance

To be controlled:

- Cell Balancing
- Charge Voltage Limits
- Limit discharge rate

Sensors required

1. Voltage & Current: INA219/INA226 (I2C-based)
   Current: ACS712/ACS758 (hall-effect, analog)
2. Temperature: DS18B20 or LM35
3. Real-time clock: DS3231 RTC module
4. BMS

To be measured:

Voltage

For 12V LFP (4 cells):
- Fully charged: 14.4–14.6 V
- Nominal: 12.8 V
- Empty (cutoff): 10–11 V

Current

Measured by Shunt resistor or Hall-effect sensor
- Positive → Charging
- Negative → Discharging

Temperature

Typical operating range:
- Ideal: 15°C – 35°C
- Max safe: ~60°C

Importance:
- High temperature → faster degradation
- Low temperature → poor charging efficiency

To be calculated:

Energy

$E = V \times Ah$

Power

$P = V \times I$

Open Circuit Voltage (OCV)

Battery voltage when no load is connected

SoC

1. Voltage-based SoC
2. Coulomb counting

$$SoC(t) = SoC(t0) - \frac{1}{C}\int I(t)dt$$

SoH

SoH = (Current Capacity / Original Capacity) × 100

DoD (Depth of Discharge)

DoD = 100 − SoC

C rating

Charge/discharge speed relative to capacity.

C = Current / Capacity

Cycle counting

Total equivalent full cycles used.
Method:

  2 half cycles = 1 full cycle
  Accumulate DoD over time

Internal resistance

New battery: ~2–5 mΩ
Aged battery: >10 mΩ

Resistance inside battery → indicator of aging.

R = ΔV / ΔI

To be graphed:

SoC vs time

Shows usage pattern and discharge rate.

SoH vs time or cycle

Shows degradation trend → critical for lifetime prediction.

Temperature vs time

Identifies overheating or poor thermal design.

Internal resistance vs time or cycle

Direct indicator of aging.

Charge-Discharge efficiency curve

η = (Energyout / Energyin) × 100

Current vs Time

Load patterns, Peak demand, Stress on battery

Power vs Time

Direct view of system demand profile

Capacity vs Cycle Number

Direct visualization of degradation curve

To be predicted:

Remaining Useful Time (RUL)

Time or cycles left before failure.
Estimated using:

- SoH trend
- Cycle count
- Temperature history

Capacity Fade Rate

How fast the battery is losing capacity.

Estimate battery lifetime

Typically,

 3000–6000 cycles
 8–15 years

Aging

Caused by:

 High temperature
 High DoD
 High C-rate

Self-discharge rate

Loss of charge without load.

Fault prediction using voltage sag

Sudden voltage drop under load → indicates:

 High internal resistance
 Weak cells

Battery Health Index to predict next maintenance

BHI = f(SoH, R, Temperature, Cycles)

To be controlled:

Cell Balancing

Ensures all 4 cells stay equal (~3.2V each).

Charge Voltage Limits

Prevents overcharging

Limit discharge rate

Prevents deep discharge damage